```
pip install pysqlite3
```

```
Collecting pysqlite3
  Downloading pysqlite3-0.6.1-cp313-cp313-manylinux_2_28_x86_64.whl.metadata (3.5 kB)
Downloading pysqlite3-0.6.1-cp313-cp313-manylinux_2_28_x86_64.whl (4.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.6/4.6 MB 32.9 MB/s eta 0:00:00
Installing collected packages: pysqlite3
Successfully installed pysqlite3-0.6.1
```

```
import sqlite3
```

```
data = sqlite3.connect('/content/data/Chinook_Sqlite.sqlite')
```

```
df = data.cursor()
```

```
df.execute("SELECT name FROM sqlite_master WHERE type = 'table';")
print(df.fetchall())
```

```
[('Album',), ('Artist',), ('Customer',), ('Employee',), ('Genre',), ('Invoice',), ('InvoiceLine',), ('MediaTy
```

```
df.execute("SELECT * FROM sqlite_master WHERE type = 'table';")
print(df.fetchall())
```

```
KEY ([MediaTypeId]) REFERENCES [MediaType] ([MediaTypeId]) \n\t\tON DELETE NO ACTION ON UPDATE NO ACTION\n)')]
```

```
df.execute("PRAGMA table_info(Employee)")
info = df.fetchall()
print(*info, sep = "\n")
```

```
(0, 'EmployeeId', 'INTEGER', 1, None, 1)
(1, 'LastName', 'NVARCHAR(20)', 1, None, 0)
(2, 'FirstName', 'NVARCHAR(20)', 1, None, 0)
```

```
(3, 'Title', 'NVARCHAR(30)', 0, None, 0)
(4, 'ReportsTo', 'INTEGER', 0, None, 0)
(5, 'BirthDate', 'DATETIME', 0, None, 0)
(6, 'HireDate', 'DATETIME', 0, None, 0)
(7, 'Address', 'NVARCHAR(70)', 0, None, 0)
(8, 'City', 'NVARCHAR(40)', 0, None, 0)
(9, 'State', 'NVARCHAR(40)', 0, None, 0)
(10, 'Country', 'NVARCHAR(40)', 0, None, 0)
(11, 'PostalCode', 'NVARCHAR(10)', 0, None, 0)
(12, 'Phone', 'NVARCHAR(24)', 0, None, 0)
(13, 'Fax', 'NVARCHAR(24)', 0, None, 0)
(14, 'Email', 'NVARCHAR(60)', 0, None, 0)
```

```
import pandas as pd
```

```
df.execute("""
  SELECT FirstName, LastName, CustomerId, Country
  FROM customer
  WHERE Country != 'USA'
""")
dff = pd.DataFrame(df.fetchall())
dff.columns = [x[0] for x in df.description]
dff.head(10)
```

	FirstName	LastName	CustomerId	Country
0	Luís	Gonçalves	1	Brazil
1	Leonie	Köhler	2	Germany
2	François	Tremblay	3	Canada
3	Bjørn	Hansen	4	Norway
4	František	Wichterlová	5	Czech Republic
5	Helena	Holý	6	Czech Republic
6	Astrid	Gruber	7	Austria
7	Daan	Peeters	8	Belgium
8	Kara	Nielsen	9	Denmark
9	Eduardo	Martins	10	Brazil

Next steps: Generate code with dff | New interactive sheet

```
df.execute("""
  SELECT Name, AlbumId, TrackId, Milliseconds
  FROM track
  WHERE Milliseconds BETWEEN 205205 AND 300000
  ORDER BY Milliseconds DESC
  LIMIT 10
""")
info = df.fetchall()
print(*info, sep = "\n")
```

```
('Breathe', 212, 2613, 299781)
('Queixa', 23, 524, 299676)
('Getaway Car', 10, 97, 299598)
('Winterlong', 201, 2485, 299389)
('Cherub Rock', 202, 2491, 299389)
```

```
('Sonata for Solo Violin: IV: Presto', 325, 3480, 299350)
('Linha Do Equador', 21, 218, 299337)
('Who Are You (Single Edit Version)', 221, 2749, 299232)
('Garden', 181, 2201, 299154)
('The Spirit Of Radio', 196, 2406, 299154)
```

```
df.execute("""
  SELECT i.billingcountry, sum(total) as 'TotalSales'
  FROM invoice AS i
  GROUP BY billingcountry
  ORDER BY TotalSales DESC
""")
info = df.fetchall()
print(*info, sep = "\n")
```

```
('USA', 523.06)
('Canada', 303.96)
('France', 195.1)
('Brazil', 190.1)
('Germany', 156.48)
('United Kingdom', 112.86)
('Czech Republic', 90.24)
('Portugal', 77.24)
('India', 75.26)
('Chile', 46.62)
('Ireland', 45.62)
('Hungary', 45.62)
('Austria', 42.62)
('Finland', 41.62)
('Netherlands', 40.62)
('Norway', 39.62)
('Sweden', 38.62)
('Spain', 37.62)
('Poland', 37.62)
('Italy', 37.62)
('Denmark', 37.62)
('Belgium', 37.62)
('Australia', 37.62)
('Argentina', 37.62)
```

```
df.execute("PRAGMA table_info(track)")
info = df.fetchall()
print(*info, sep = "\n")
```

```
(0, 'TrackId', 'INTEGER', 1, None, 1)
(1, 'Name', 'NVARCHAR(200)', 1, None, 0)
(2, 'AlbumId', 'INTEGER', 0, None, 0)
(3, 'MediaTypeId', 'INTEGER', 1, None, 0)
(4, 'GenreId', 'INTEGER', 0, None, 0)
(5, 'Composer', 'NVARCHAR(220)', 0, None, 0)
(6, 'Milliseconds', 'INTEGER', 1, None, 0)
(7, 'Bytes', 'INTEGER', 0, None, 0)
(8, 'UnitPrice', 'NUMERIC(10,2)', 1, None, 0)
```

```
df.execute("""
  SELECT * FROM track
""")
info = df.fetchall()
print(*info, sep = "\n")
```

Show hidden output

```
df.execute("""
  SELECT t.Name, SUM(il.Quantity) AS TotalSold
  FROM InvoiceLine il
  JOIN Track t ON il.TrackId = t.TrackId
  GROUP BY t.TrackId
  ORDER BY TotalSold DESC
  LIMIT 10
""")
info = df.fetchall()
print(*info, sep = "\n")
```

```
('Balls to the Wall', 2)
('Inject The Venom', 2)
('Snowballed', 2)
('Overdose', 2)
```

```
('Deuces Are Wild', 2)
('Not The Doctor', 2)
('Por Causa De Você', 2)
('Welcome Home (Sanitarium)', 2)
('Snowblind', 2)
('Cornucopia', 2)
```

```
df.execute("""
  SELECT BillingCountry, SUM(Total) AS TotalRevenue
  FROM Invoice
  GROUP BY BillingCountry
  ORDER BY TotalRevenue DESC
  LIMIT 1
""")
print(df.fetchone())
```

```
('USA', 523.06)
```

```
df.execute("""
  SELECT e.FirstName, e.LastName, SUM(i.Total) AS TotalSales
  FROM Employee e
  JOIN Customer c ON e.EmployeeId = c.SupportRepId
  JOIN Invoice i ON c.CustomerId = i.CustomerId
  GROUP BY e.EmployeeId
  ORDER BY TotalSales DESC
  LIMIT 1
""")
print(df.fetchone())
```

```
('Jane', 'Peacock', 833.04)
```

```
query_df = pd.read_sql_query("""
  SELECT e.FirstName, e.LastName, SUM(i.Total) AS TotalSales
  FROM Employee e
  JOIN Customer c ON e.EmployeeId = c.SupportRepId
  JOIN Invoice i ON c.CustomerId = i.CustomerId
```

```
  GROUP BY e.EmployeeId
  ORDER BY TotalSales DESC
""", data)

display(query_df)
```

	FirstName	LastName	TotalSales
0	Jane	Peacock	833.04
1	Margaret	Park	775.40
2	Steve	Johnson	720.16

Next steps: Generate code with query_df | New interactive sheet

```
import matplotlib.pyplot as plt
import seaborn as sns

query_df['FullName'] = query_df['FirstName'] + ' ' + query_df[

sns.set_theme(style="whitegrid")
fig, ax = plt.subplots(figsize=(8, 5))

sns.barplot(
    x='TotalSales',
    y='FullName',
    data=query_df,
    palette='viridis',
    ax=ax
)

ax.set_title('Total Sales Revenue by Sales Employee', fontsize=
ax.set_xlabel('Total Sales ($)', fontsize=12)
ax.set_ylabel('Employee Name', fontsize=12)

for index, value in enumerate(query_df['TotalSales']):
```

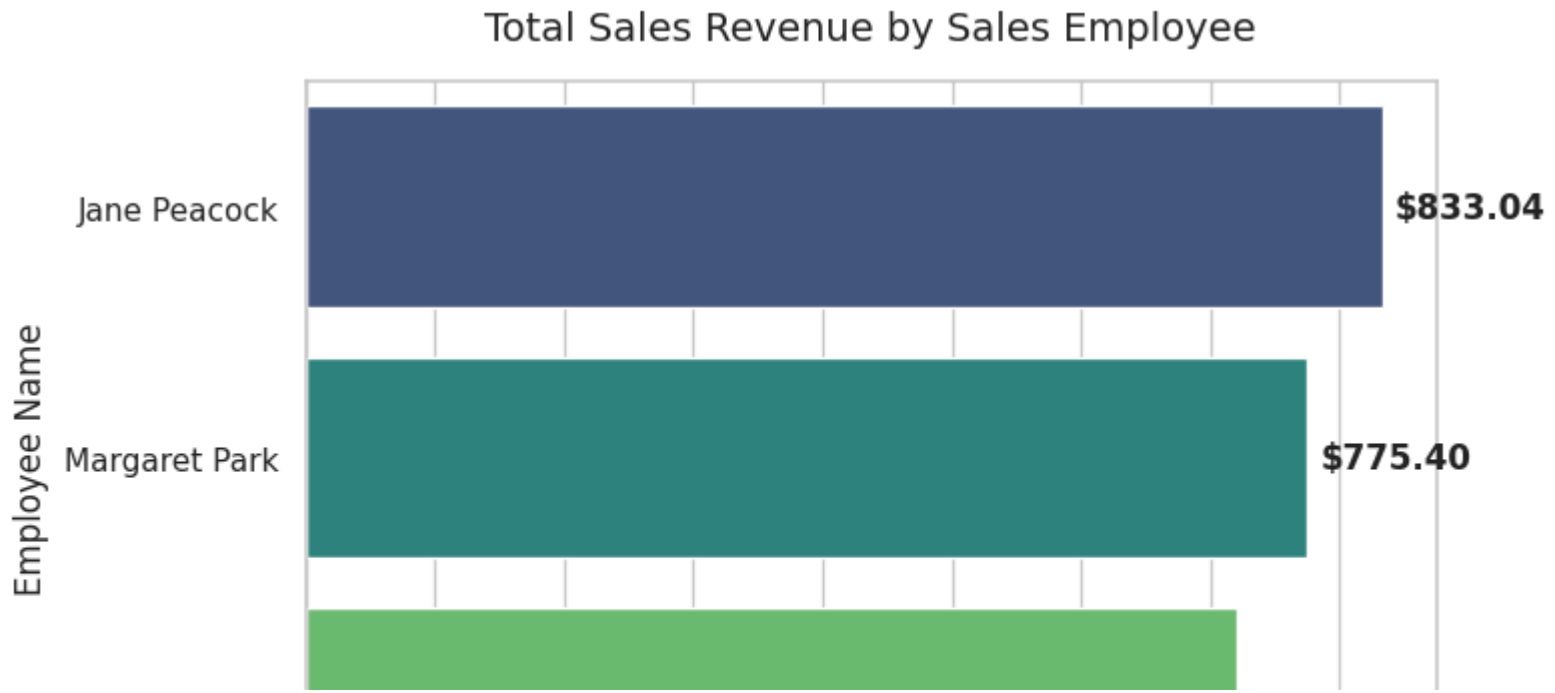
```
    ax.text(value + 10, index, f"${value:,.2f}", va='center', f

plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_2003/2979722020.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variab

  sns.barplot(
```




```
Start coding or generate with AI.
```